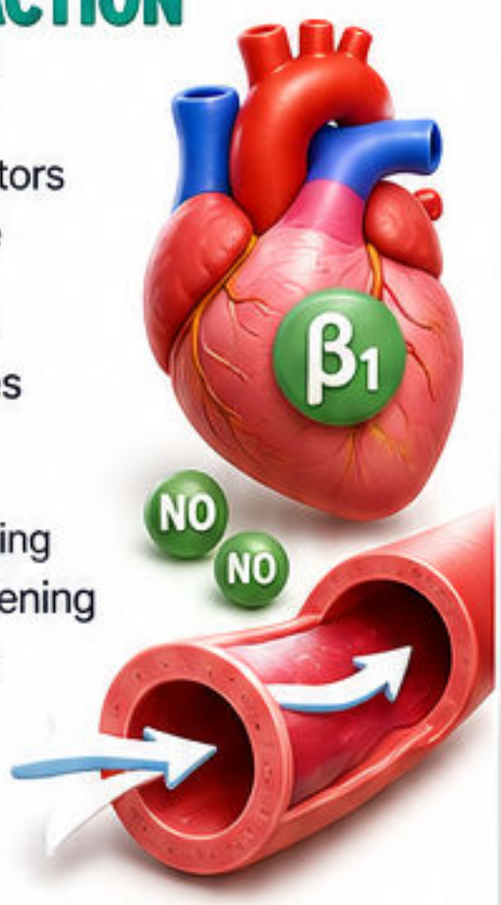
NEBIVOLOL [BYSTOLIC]

Pharma Talk

Class: Antihypertensive + Cardioselective beta-1 blocker

1 MECHANISM OF ACTION (MOA) & TARGET



- **TARGET:** Beta-1 adrenergic receptors & endothelial nitric oxide synthase
- **MOA:** Blocks beta-1 ➡ ⬇ HR & contractility ➡ ⬇ BP | Stimulates NO ➡ vasodilation ➡ ⬇ PVR
- **Analogy:** A dual-action brake slowing the heart while simultaneously widening the downstream vascular pipes to reduce overall workload.

2 FORMS & STRENGTHS

- **Tablets:** 2.5 mg, 5 mg, 10 mg



3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)

- **FDA:** Hypertension management in adult patients to reduce the risk of fatal and nonfatal cardiovascular events
- **Off-Label:** Heart failure in elderly patients, chronic stable angina pectoris, rate control in atrial fibrillation



4 DOSAGE & ADMINISTRATION

A) ADULT DOSE

- **Standard:** 5 mg once daily
- **Max:** 40 mg once daily
- **Route:** PO
- **Frequency:** Once daily
- Take with or without food consistently

B) PEDIATRIC DOSE

PEDIATRIC USE – Safety/efficacy not established.

C) ORGAN ADJUSTMENT

- **Renal:** CrCl <30 mL/min: 2.5 mg daily
- **Hepatic:** Child-Pugh moderate/severe: 2.5 mg daily

D) ADMINISTRATION

- **Route:** PO
- Take with or without food consistently
- Take at the same time each day

5 PHARMACOKINETICS & DYNAMICS (PK/PD)



- **Onset:** 1–2 weeks for full effect
- **$T_{1/2}$:** 12–19 hrs
- **Metabolism:** Extensive hepatic metabolism via CYP2D6

6 ADVERSE EFFECTS & TOXICITY



- **Common:** Headache | Fatigue | Dizziness | Nausea | Diarrhea | Insomnia | Peripheral edema
- **Severe:** Bradycardia, heart block, bronchospasm, masked hypoglycemia symptoms



7 CONTRAINDICATIONS & INTERACTIONS

ABSOLUTE CONTRAINDICATIONS

- **Severe** bradycardia
- Cardiogenic shock

BOXED WARNING

Ischemic heart disease **exacerbation upon** abrupt withdrawal

MAJOR DRUG INTERACTIONS

- CYP2D6 inhibitors ➡ ⬇ metabolism ➡ ⬆ toxicity
- Digoxin ➡ synergistic AV node suppression ➡ profound bradycardia
- NSAIDs ➡ ⬇ antihypertensive efficacy



8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)

- **Pregnancy:** Fetal harm risk
- **Lactation:** Use caution; unknown if excreted in milk
- **Geriatric:** Start 2.5 mg with careful monitoring

9 CLINICAL PEARLS



- Highly cardioselective; safer in mild asthma or COPD.
- NO-mediated vasodilation preserves erectile function better than older beta-blockers.
- CYP2D6 poor metabolizers require slower dose titration.
- Taper over 1–2 weeks to avoid rebound tachycardia.
- Monitor HR closely in elderly patients starting therapy.

10 GLOSSARY OF ABBREVIATIONS

- **HR** = Heart rate
- **BP** = Blood pressure
- **NO** = Nitric oxide
- **PVR** = Peripheral vascular resistance
- **CrCl** = Creatinine clearance
- **mL/min** = Milliliters per minute
- **CYP2D6** = Cytochrome P450 2D6
- **AV** = Atrioventricular
- **COPD** = Chronic obstructive pulmonary disease